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LongSplat: Robust Unposed 3D Gaussian Splatting for Casual Long Videos

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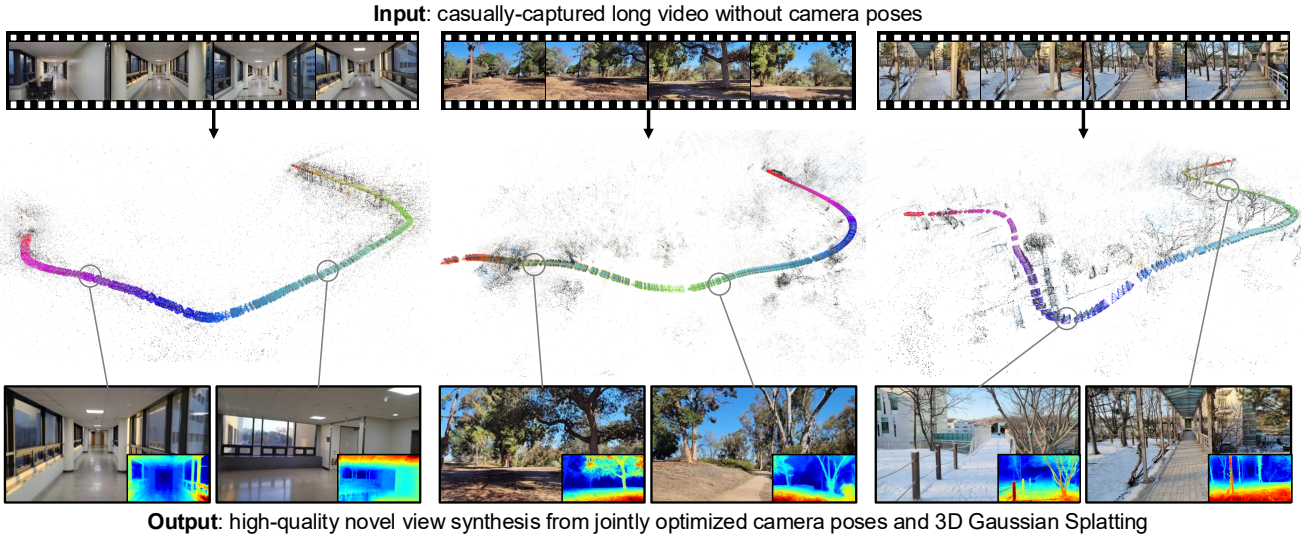


Figure 1. **LongSplat achieves robust novel view synthesis from casually captured long videos without provided camera poses.** Our approach jointly optimizes camera poses and 3D Gaussian Splatting, producing accurate and visually coherent reconstructions even under challenging conditions.

Abstract

LongSplat addresses critical challenges in novel view synthesis (NVS) from casually captured long videos characterized by irregular camera motion, unknown camera poses, and expansive scenes. Current methods often suffer from pose drift, inaccurate geometry initialization, and severe memory limitations. To address these issues, we introduce LongSplat, a robust unposed 3D Gaussian Splatting framework featuring: (1) Incremental Joint Optimization that concurrently optimizes camera poses and 3D Gaussians to avoid local minima and ensure global consistency; (2) a robust Pose Estimation Module leveraging learned 3D priors; and (3) an efficient Octree Anchor Formation mechanism that converts dense point clouds into anchors based on spatial density. Extensive experiments on challenging benchmarks demonstrate that LongSplat achieves state-of-the-art results, substantially improving rendering quality, pose accuracy, and computational efficiency compared to prior approaches. Project page: https://linjohnss.github.io/longsplat/

1. Introduction

High-quality 3D reconstruction and novel view synthesis (NVS) are essential for applications such as virtual reality, augmented reality, virtual tourism, and cultural heritage preservation. They also play a crucial role in video editing tasks like stabilization, visual effects, and digital mapping for real estate or pedestrian-level navigation. With the widespread availability of smartphones and action cameras, casually captured videos have emerged as a significant source of 3D content. Unlike professionally acquired datasets, casual videos present challenging characteristics: irregular camera trajectories, long sequences spanning hundreds or thousands of frames, and the absence of reliable camera poses or precise geometric priors.

Addressing novel view synthesis (NVS) from casually captured videos poses two critical challenges: robust camera pose estimation over extended trajectories and efficient representation of large-scale scenes. Traditional methods rely on accurate poses from Structure-from-Motion (SfM) preprocessing, yet as shown in Fig. 2, pipelines like COLMAP [50]

frequently fail in casual settings. COLMAP-free methods, such as CF-3DGS [14], often encounter severe memory constraints, limiting their effectiveness for large-scale scenarios. Similarly, methods like LocalRF [39] struggle with complex camera trajectories, resulting in fragmented reconstructions. Foundation models like MASt3R [27] provide fast initial estimates but suffer inaccuracies and drift in long videos, severely affecting reconstruction quality.

To address these limitations, we introduce **LongSplat**, a robust unposed 3D Gaussian Splatting (3DGS) [22] framework designed specifically for casual long videos. As illustrated in Fig. 1, LongSplat achieves accurate novel view synthesis without relying on provided camera poses. LongSplat departs from traditional pipelines by jointly optimizing camera poses and 3DGS in a unified framework. It integrates a correspondence-guided Pose Estimation Module with 3DGS geometry and photometric refinements to improve pose accuracy, even under large-scale and unstructured camera motion. Furthermore, an efficient Octree Anchor Formation mechanism converts dense point clouds into anchors based on spatial density, significantly reducing memory usage while retaining detailed scene structures. These components work together in an incremental joint optimization strategy that avoids local minima and ensures global geometric consistency across extensive sequences.

Extensive experiments on challenging datasets, including Tanks and Temples, Free, and Hike datasets, demonstrate that LongSplat consistently outperforms existing approaches, significantly improving rendering quality and pose accuracy. Compared to conventional methods shown in Fig. 2, LongSplat produces clearer and more coherent reconstructions, effectively addressing pose drift and memory limitations and substantially advancing the state-of-the-art. The main contributions of our work are:

- An incremental joint optimization approach for simultaneous camera pose and 3DGS reconstruction, reducing local minima and ensuring global consistency.
- A robust pose estimation module leveraging learned 3D priors for accurate camera pose estimation.
- An adaptive Octree Anchor Formation strategy that significantly reduces memory usage while preserving reconstruction quality.

2. Related Work

Novel View Synthesis. Novel View Synthesis (NVS) generates new perspectives from captured images, evolving from early pixel interpolation methods [8] to depth-based warping techniques [28] and 3D reconstruction-based rendering [6, 12]. Various representations have been explored, including planes [17, 18], meshes [20, 47, 48], point clouds [67, 74], and multi-plane images [29, 57, 76]. Neural Radiance Fields (NeRF) [40] revolutionized photorealistic rendering, with subsequent improvements in anti-

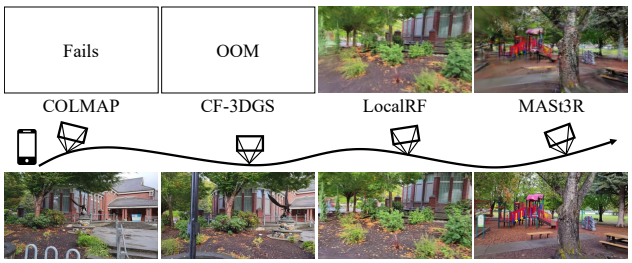


Figure 2. **Novel view synthesis for casual long videos.** Existing methods encounter significant challenges when reconstructing scenes from casually captured long videos: COLMAP [50] fails due to incorrect camera pose estimation, CF-3DGS [14] suffers from out-of-memory issues, LocalRF [39] struggles with complex trajectories, and MASt3R [27]+Scaffold-GS [36] provides inaccurate poses leading to degraded rendering quality. In contrast, LongSplat robustly handles these challenges, yielding accurate camera poses and high-quality novel view synthesis without memory constraints.

aliasing [2–4, 73], reflectance [1, 58], sparse view training [24, 43, 66, 68], faster training [41, 45, 49], and rendering speed [15, 34, 53, 71]. Recent works have extended NeRF to few-shot scenarios without learned priors [32], domain-specific applications such as autonomous driving environments [51], and dynamic scenes with human pose variations [38]. Point-based methods [22, 37, 67, 74], particularly 3D Gaussian Splatting (3DGS) [22], enable real-time rendering through explicit representations. Recent advances have extended 3DGS capabilities to dynamic specular scenes with physically-based rendering [13], developed compression techniques for efficient storage and transmission [72], and improved robustness for unconstrained image scenarios [19]. However, most approaches still rely on pre-computed camera parameters from SfM [16, 32, 42, 50, 55].

Unposed Novel View Synthesis. Recent work has aimed to eliminate camera estimation preprocessing. i-NeRF [70] predicts camera poses using pre-trained NeRF. NeRFmm [64] jointly optimizes NeRF and camera poses for forward-facing scenes, with SiNeRF [65] offering improvements. BARF [31] and GARF [10] address gradient inconsistency through coarse-to-fine positional encoding but require good initialization. Advanced approaches [5, 9, 35, 39] leverage pre-trained networks for geometric priors, with NoPe-NeRF [5] incorporating monocular depth priors and CF-3DGS [14] using progressive optimization. Recent methods have improved robustness in joint optimization of camera poses and scene geometry using decomposed low-rank tensorial representations [7] and dynamic radiance fields [35]. These methods typically assume small pose perturbations [10, 31], limited camera motion [64, 65], or additional priors [5, 11, 21, 39], struggling with challenging trajectories, like Free dataset[26, 44].

Large-scale Novel View Synthesis. Extending NVS to large-scale environments introduces memory and computational

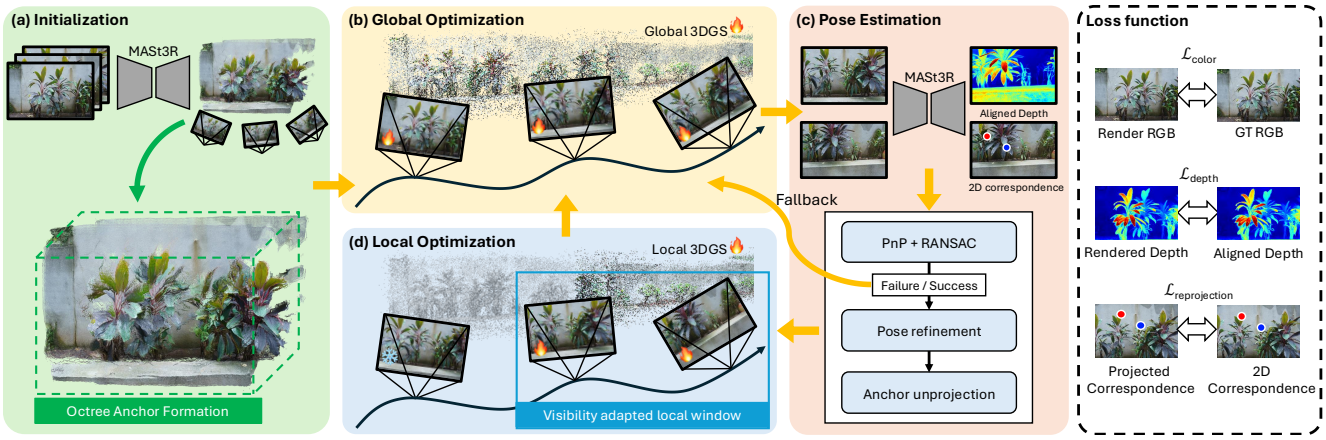


Figure 3. **Overview of the LongSplat framework.** Given a casually captured long video without known poses, LongSplat incrementally reconstructs the scene through tightly coupled pose estimation and 3D Gaussian Splatting. (a) Initialization converts MASt3R [27] global aligned point cloud into an octree-anchored 3DGS. (b) Global Optimization jointly refines all camera poses and 3D Gaussians for global consistency. (c) Pose estimation estimates each new frame pose via correspondence-guided PnP, applies photometric refinement, and updates octree anchors using unprojected points. If PnP fails, a fallback triggers global re-optimization to recover. (d) Incremental Optimization alternates between Local Optimization within a visibility-adapted window and periodic Global Optimization to propagate consistent updates across frames. (e) All optimization stages leverage a unified objective composed of photometric loss, depth loss, and reprojection loss to ensure accurate geometry and appearance reconstruction.

challenges that NeRF's implicit global representation struggles with. Recent research employs scene partitioning strategies for managing large scenes [23, 54, 56]. Progressive optimization techniques have been developed for robust view synthesis in large-scale scenes from casually captured videos [39]. At the same time, MVS-based approaches have been enhanced to handle generalizable view synthesis at scale [52]. For indoor environments, methods like GenRC [30] enable room-scale 3D reconstruction from sparse image collections. 3DGS offers explicit representation advantages through Gaussian primitive. VastGaussian [33] divides scenes into separately optimized blocks[22]. Scaffold-GS [36] introduces anchor-based Gaussian representation with fixed-resolution grids, though it requires SfM initialization. Octree-GS [46] implements fixed-level octrees with preset resolutions but similarly depends on SfM. Unlike these approaches, our method dynamically adjusts voxel size based on point cloud density, without dependency on SfM, and addresses unposed, large-scale, casually captured videos through adaptive Octree Anchor Formation.

Casual Long Videos. Casual long videos present unique challenges: free-moving trajectories, lack of pose information, and continuously expanding scenes. LocalRF [39] addresses these through progressive localized field construction but suffers from slow training and fragmentation under irregular camera movements. 3D Foundation Models [59], including DUSt3R [62], MASt3R [27], Fast3r [69], and CUT3R [61], estimate poses and geometry directly but accumulate errors in long sequences. LongSplat treats foundation model outputs as soft priors, jointly optimizing them with 3D Gaussian Splatting while progressively correcting poses and geometry through combined PnP and optimization strategies.

3. Method

LongSplat reconstructs long video sequences with unknown camera poses and unconstrained trajectories through a fully incremental pipeline based on octree-anchored 3D Gaussian Splatting. The process begins with octree anchor formation, where per-frame dense point clouds are structured into an adaptive representation. Next, camera poses are estimated and refined using correspondence-guided initialization and photometric alignment. Finally, the reconstruction alternates between local optimization, which updates Gaussians within a visibility-adapted window, and global refinement, which ensures long-term consistency. This design allows LongSplat to robustly handle long, unconstrained trajectories while adapting to scene complexity and minimizing drift.

3.1. Preliminaries

Gaussian Splatting. 3D Gaussian Splatting (3DGS) [22] represents the scene as a set of 3D Gaussians, each defined by a center $\mu \in \mathbb{R}^3$, a covariance matrix Σ, color, scale, rotation, and opacity. The covariance is factorized into a rotation $R \in SO(3)$ and a diagonal scale matrix S, giving:

$$G(x) = e^{-\frac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)}, \quad \Sigma = RSS^\top R^\top. \tag{1}$$

This parameterization allows each Gaussian to adaptively capture local scene geometry.

To render the scene, each Gaussian is projected onto the image plane using the camera pose W, resulting in a 2D Gaussian with covariance $\Sigma_{2D} = JW\Sigma W^\top J^\top$, where J is the Jacobian of the projective transformation. The final rendered color and depth are computed via alpha blending:

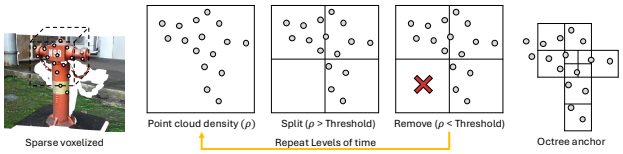


Figure 4. **Visualization of our proposed Octree Anchor Formation strategy.** Given an initial sparse voxelized point cloud, we iteratively perform density-guided adaptive voxel splitting and pruning. Voxels with point cloud density (ρ) exceeding a threshold are split, while those with density below the threshold are pruned. Repeated across multiple octree levels, this adaptive octree anchor design significantly reduces memory usage, allowing efficient representation and rendering of large-scale scenes.

$$C = \sum_{i=1}^{N} c_i \, \alpha_i \prod_{j=1}^{i-1} (1 - \alpha_j), \quad D = \sum_{i=1}^{N} d_i \, \alpha_i \prod_{j=1}^{i-1} (1 - \alpha_j), \tag{2}$$

where c_i and α_i denote the color and opacity of the i-th Gaussian, respectively. d_i denotes the depth value along the ray at the Gaussian's center.

Anchor-based 3D Gaussian Splatting. To enhance memory efficiency and robustness in large scenes, Scaffold-GS [36] introduces the anchor-based 3DGS representation. Instead of directly maintaining individual Gaussians, the scene is first divided into sparse voxels, each acting as an anchor. From each anchor, k Gaussians are initialized with positions relative to the anchor center:

$$\{\mu_0, \mu_1, \ldots, \mu_{k-1}\} = x_v + \{O_0, O_1, \ldots, O_{k-1}\} \cdot l_v, \tag{3}$$

where x_v denotes the anchor position, $\{O_i\}$ are offset vectors, and l_v is a scaling factor. Each Gaussian's opacity, rotation, scale, and color are decoded from an anchor feature through lightweight MLPs. For opacity, the formulation is:

$$\{\alpha_0, \alpha_1, \ldots, \alpha_{k-1}\} = F_\alpha(\hat{f}_v, \Delta v_c, \hat{d}_v), \tag{4}$$

where F_α is an MLP taking the anchor feature $\hat{f}_v$, the relative view distance Δv_c, and the view direction $\hat{d}_v$ as inputs.

Anchor Initialization. In traditional Scaffold-GS, initial anchors are derived from sparse SfM point clouds. Points are voxelized to form anchor centers:

$$V = \{v \mid v = \lfloor \frac{p}{\epsilon} \rfloor \cdot \epsilon, \forall p \in P\}, \tag{5}$$

where P is the SfM point cloud and ϵ is the voxel size. Each anchor holds a local feature, managing its associated Gaussians. This design ensures structured densification and pruning, adapting Gaussian density to scene complexity and improving both memory and rendering efficiency.

3.2. Octree Anchor Formation

In large-scale casual video settings, memory efficiency and scene adaptability are essential. Our Octree Anchor Formation dynamically adjusts spatial resolution based on observed geometry, enabling scalable and redundant-free anchor management. LongSplat constructs structured anchors from MASt3R's per-frame dense point clouds using an adaptive octree (Fig. 3 (a)). Unlike Scaffold-GS, which relies on a fixed-resolution grid, we progressively subdivide space based on local point density. Each point cloud $\mathbf{P} = \{p_i\}$ is voxelized into a sparse grid at resolution ϵ_0. Voxels exceeding a density threshold τ_{split} split into 8 smaller voxels:

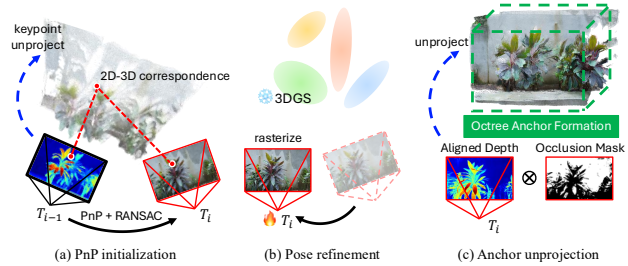
$$\epsilon_{l+1} = \frac{1}{2}\epsilon_l. \tag{6}$$

This process repeats up to a maximum level L. Low-density voxels (density $\rho_v < \tau_{\text{prune}}$) are removed to reduce redundancy (Fig. 4).

Each anchor inherits a spatial scale proportional to its voxel size, ensuring coarse anchors for sparsely observed areas and finer anchors for detailed regions:

$$s_v \propto \epsilon_v. \tag{7}$$

To further prevent unnecessary duplication, newly generated anchors are compared to existing ones. If significant spatial overlap exists, the new anchor is discarded. This density-adaptive, duplication-free octree formation ensures compact memory usage while preserving adaptive resolution across scenes.



(a) PnP initialization (b) Pose refinement (c) Anchor unprojection

Figure 5. **Detailed illustration of our camera pose estimation.** (a) PnP initialization: Given correspondences between the predicted 3D anchor points from frame T_{i-1} and the 2D keypoints detected in frame T_i we employ PnP with RANSAC to robustly estimate an initial camera pose. (b) Pose refinement: The estimated pose is further refined by rasterizing the 3DGS scene and iteratively minimizing reprojection error to enhance pose accuracy. (c) Anchor unprojection: Newly observed regions are detected via an occlusion mask, computed by forward-warping the previous frame's rendered depth. These regions are unprojected into 3D and converted into anchors via Octree Anchor Formation.

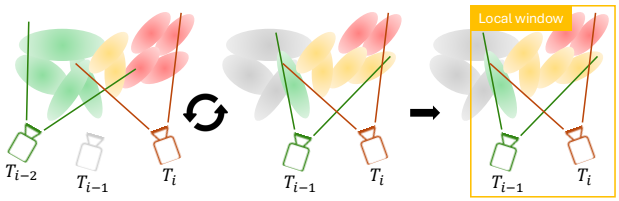
3.3. Pose Estimation module

Accurate and robust camera pose estimation is essential for consistent reconstruction in unposed long video settings. We estimate each pose using 2D-3D correspondences derived from MASt3R, followed by photometric refinement against the current 3D Gaussian scene to maintain coherence across evolving 3D structures (Fig. 3 (c)).

For each new frame t, MASt3R provides 2D correspondences $\{(x_i, x_i')\}$ between frame t and $t-1$, allowing back-projection of matched points x_i to 3D via:

$$X_i = D_{t-1}(x_i) \cdot K^{-1} \tilde{x}_i. \tag{8}$$

These 2D-3D correspondences $\{(x_i', X_i)\}$ are used to solve the initial pose T_t via PnP (Fig. 5 (a)), followed by photometric refinement that minimizes (Fig. 5 (b)):



(a) IOU < Threshold (b) IOU > Threshold (c) Local optimize

Figure 6. **Illustration of our Visibility-Adapted Local Window strategy for local optimization.** To ensure balanced training of the 3D Gaussians, we dynamically define the optimization window based on anchor visibility overlap. Specifically, we compute the Intersection-over-Union (IoU) of visible anchors between consecutive views. Suppose the visibility IoU is below a certain threshold (a). In that case, the local optimization window is adjusted by removing the earliest frame, iteratively repeating until a suitable window with IoU above the threshold is found (b). This approach ensures balanced training coverage and enhances local reconstruction details during optimization (c).

$$\mathcal{L}_{\text{photo}} = \sum_{p \in \Omega} \|I_t(p) - \hat{I}_t(p)\|^2, \tag{9}$$

where I_t is the observed frame and $\hat{I}_t$ is the rendering using the current 3DGS. This ensures the pose aligns with the evolving scene.

To correct MASt3R's depth scale drift, we compute a scale factor $\hat{s}_t$ by comparing the rendered depth D_{t-1} and MASt3R's aligned depth D_t^{align}:

$$\hat{s}_t = \frac{\langle D_{t-1}, D_t^{\text{align}} \rangle}{\langle D_t^{\text{align}}, D_t^{\text{align}} \rangle}. \tag{10}$$

This rescaled depth ensures consistent scale across frames.

As the camera moves, newly visible regions are detected via an occlusion mask M_{occ}, derived by forward-warping D_{t-1} to frame t and comparing it to the rescaled depth D_t^{MASt3R} (Fig. 5 (c)). Newly visible pixels are unprojected into 3D using:

$$p_i = D_{t,\mathbf{u}_i}^{\text{MASt3R}} \cdot \mathbf{K}^{-1} \mathbf{u}_i. \tag{11}$$

These new points are converted into octree anchors using the *Octree Anchor Formation* described in Sec. 3.2, with overlapping anchors removed to avoid redundancy (Fig. 5 (c)). This process incrementally expands the scene while maintaining structural regularity.

3.4. Incremental Joint Optimization

To handle casually captured long videos, LongSplat adopts a progressive incremental optimization framework that alternates between per-frame local reconstruction and cross-frame global consistency refinement.

Initialization. We begin with a small set of initial frames. Camera poses and dense point clouds for these frames are estimated using MASt3R [27], followed by converting the point cloud into an initial octree-anchored 3DGS using the proposed Octree Anchor Formation (Fig. 3 (a). When camera intrinsics are unavailable, we directly adopt MASt3R's estimated focal length.

Global Optimization. After initialization, we jointly optimize all 3D Gaussian parameters and camera poses across all processed frames (Fig. 3 (b)). This global optimization ensures geometric consistency across the entire sequence, reducing accumulated pose drift and local misalignments.

Frame Insertion and Pose Estimation. As new frames arrive, we estimate their poses using the correspondence-guided PnP initialization and refinement strategy described in Sec. 3.3. If PnP fails due to insufficient feature correspondences or poor initialization, we trigger a fallback mechanism that re-optimizes all past frames globally before retrying pose estimation. This iterative fallback enhances robustness under challenging motion or weak texture (Fig. 3 (c)).

Local Optimization with Visibility-Adaptive Window. Once the pose is estimated, we optimize only the Gaussians visible in the new frame's frustum, while constraining them with observations from nearby frames in a dynamically selected *visibility-adapted local window* (Fig. 6). Covisibility between frames is measured by:

$$\text{IoU}(t, t') = \frac{|\mathcal{V}(t) \cap \mathcal{V}(t')|}{|\mathcal{V}(t) \cup \mathcal{V}(t')|}, \tag{12}$$

where $\mathcal{V}(t)$ denotes the set of Gaussians visible in frame t. Frames with covisibility below a threshold τ are excluded from the window. This adaptive mechanism ensures local Gaussians are consistently supervised by reliable multi-view constraints, balancing efficiency and accuracy.

Final Global Refinement. In the final step, a final global refinement jointly optimizes all Gaussians and camera poses over the sequence. This final pass further improves both rendering quality and long-range pose consistency.

Depth and Reprojection Losses. To provide additional supervision in newly revealed regions, where multi-view observations are insufficient, we introduce two regularization terms. A monocular depth loss encourages rendered depth to match MASt3R's scale-aligned depth prior:

$$\mathcal{L}_{\text{depth}} = \|D^{\text{rendered}} - D^{\text{MASt3R}}\|^2. \tag{13}$$

Additionally, a keypoint reprojection loss enforces alignment between projected 3D keypoints and their 2D observations:

$$\mathcal{L}_{\text{reprojection}} = \sum_k \|\pi(\mathbf{X}_k) - \mathbf{u}_k\|^2, \tag{14}$$

where $\pi(\cdot)$ denotes projection using the current pose.

Total Loss. Throughout the entire incremental reconstruction pipeline, each processed frame is optimized using the following objective:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{photo}} + \lambda_{\text{depth}} \mathcal{L}_{\text{depth}} + \lambda_{\text{reprojection}} \mathcal{L}_{\text{reprojection}}, \tag{15}$$

This combined loss applies to both local and global optimization stages, ensuring coherent multi-view, robust pose refinement, and stable geometry reconstruction across the evolving scene.

Table 1. **Quantitative comparison on the Free dataset [60] across various baseline methods.** Methods such as CF-3DGS [14] frequently encounter out-of-memory issues, denoted by "-". Our method consistently outperforms all baselines across diverse scenes, delivering superior rendering quality and robustness, especially in challenging environments characterized by complex camera trajectories and varied geometric structures. "*": Initialized with MASt3R poses, then jointly optimized.

Scenes	COLMAP [50] + F2-NeRF [60]			COLMAP [50] + Scaffold-GS [36]			MASt3R [27] + Scaffold-GS [36]			MASt3R [27] + Scaffold-GS [36]*			CF-3DGS [14]			NoPe-NeRF [5]			LocalRF [39]			Ours		
	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
Grass	23.44	0.58	0.45	26.75	0.82	0.20	22.65	0.61	0.34	25.06	0.79	0.21	-	-	-	16.39	0.27	0.81	18.84	0.35	0.60	**26.16**	**0.80**	**0.22**
Hydrant	23.75	0.74	0.28	26.66	0.86	0.12	23.22	0.71	0.21	25.68	0.83	0.12	-	-	-	17.94	0.43	0.66	19.19	0.48	0.48	**24.69**	**0.79**	**0.18**
Lab	24.34	0.83	0.26	28.27	0.92	0.10	20.66	0.74	0.25	22.42	0.80	0.18	-	-	-	17.42	0.52	0.63	17.22	0.55	0.47	**27.11**	**0.87**	**0.15**
Pillar	28.05	0.79	0.23	31.75	0.90	0.12	23.95	0.70	0.28	22.88	0.67	0.24	14.55	0.40	0.66	18.88	0.44	0.75	22.98	0.59	0.49	**30.44**	**0.88**	**0.16**
Road	26.03	0.80	0.27	30.45	0.92	0.10	24.23	0.73	0.25	25.05	0.78	0.27	-	-	-	17.48	0.44	0.79	20.68	0.54	0.56	**27.73**	**0.84**	**0.20**
Sky	25.10	0.86	0.24	28.34	0.92	0.12	23.26	0.80	0.22	25.37	0.88	0.14	-	-	-	16.18	0.51	0.65	18.76	0.60	0.46	**28.07**	**0.91**	**0.13**
Stair	28.14	0.84	0.22	32.13	0.93	0.10	23.35	0.71	0.30	24.46	0.79	0.28	13.41	0.41	0.63	19.14	0.47	0.69	23.55	0.66	0.38	**31.00**	**0.89**	**0.16**
Avg.	25.55	0.78	0.28	29.19	0.90	0.12	23.05	0.72	0.27	24.42	0.79	0.21	13.98	0.41	0.65	17.63	0.44	0.71	20.17	0.54	0.49	**27.88**	**0.85**	**0.17**

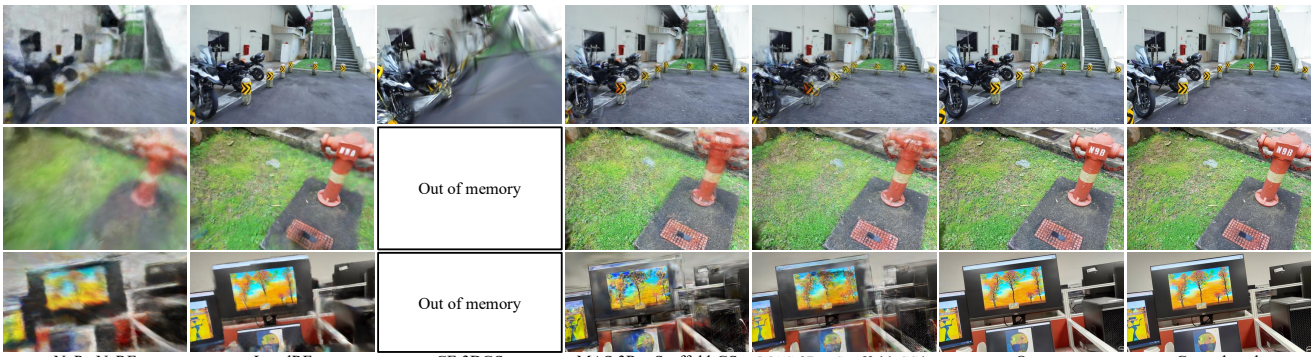


Figure 7. **Qualitative comparison on the Free dataset [60].** We compare our method with state-of-the-art approaches including NoPe-NeRF [5], LocalRF [39], CF-3DGS [14], and MASt3R [27] combined with Scaffold-GS [36]. CF-3DGS fails due to memory constraints (OOM), and other baseline methods exhibit artifacts or blurry reconstructions. In contrast, our method produces results closest to the ground truth, demonstrating clearer details, accurate geometry, and visually consistent rendering, particularly under challenging scene structures and complex camera trajectories. "*": Initialized with MASt3R poses, then jointly optimized.

4. Experiments

4.1. Experimental Setup

Datasets. We evaluate LongSplat on three challenging real-world datasets with varying difficulty levels:

- **Tanks and Temples [25] (Standard)**: Eight scenes with smooth, forward-facing camera trajectories, evaluated at full resolution. Every 8th frame is used for testing.
- **Free Dataset [60] (Moderate)**: Seven handheld videos featuring complex, unconstrained trajectories with multiple foreground objects, evaluated at 1/2 resolution. Frequent scene changes make memory-efficient 3D representation essential. Every 8th frame is tested.
- **Hike Dataset [39] (Hard)**: Long videos with hundreds to thousands of frames, complex trajectories, and detailed geometry, evaluated at 1/4 resolution. The scale and duration demand adaptive memory management. Every 10th frame is used for testing.

Evaluation Metrics. We evaluate novel view synthesis quality using PSNR, SSIM [63], and LPIPS [75]. Pose accuracy is measured with Absolute Trajectory Error (ATE) and Relative Pose Error (RPE), using COLMAP poses as ground truth. We also report model size, training time, and FPS to assess computational efficiency.

Table 2. **Quantitative evaluation of camera pose estimation accuracy on the Free dataset [60].** Our method achieves superior performance across most scenes, significantly reducing pose errors compared to state-of-the-art approaches. "*": Initialized with MASt3R poses, then jointly optimized.

Method	RPE$_t$↓	RPE$_r$↓	ATE↓
MASt3R [27] + Scaffold-GS [36]	0.162	0.265	0.013
MASt3R [27] + Scaffold-GS [36]*	0.083	0.176	0.008
CF-3DGS [14]	0.234	3.442	0.022
NoPe-NeRF [5]	6.231	4.822	0.576
LocalRF [39]	0.754	7.086	0.035
Ours	**0.028**	**0.103**	**0.004**

Baselines. We compare LongSplat with COLMAP-based methods (COLMAP [50]+F2-NeRF [60] / 3DGS [22] / Scaffold-GS [36]) and unposed methods (NoPe-NeRF [5], LocalRF [39], CF-3DGS [14]). Additionally, we evaluate a naïve baseline combining MASt3R's [27] predicted point cloud and poses with Scaffold-GS. During training, camera poses are either fixed (MASt3R + Scaffold-GS) or jointly optimized (MASt3R + Scaffold-GS*).

Implementation Details. We implement LongSplat based on Scaffold-GS [36], using its learning rate schedule and growing/pruning rules. Each anchor emits k Gaussians predicted by a lightweight 2-layer MLP. The initial sparse voxel

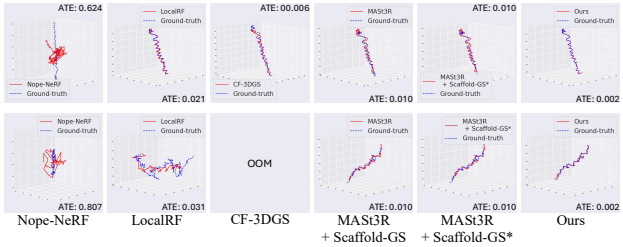


Figure 8. **Visualization of camera trajectories on Free dataset [60].** CF-3DGS [14] encounters OOM and fails for long sequences, whereas our method reliably estimates accurate, stable trajectories, demonstrating superior robustness.

Table 3. **Quantitative evaluation of novel view synthesis quality on the Tanks and Temples dataset [25].** Our proposed LongSplat consistently surpasses existing methods across multiple scenes.

Method	PSNR↑	SSIM↑	LPIPS↓	RPE_t↓	RPE_r↓	ATE↓
COLMAP+3DGS [22]	30.21	0.92	0.10	–	–	–
MASt3R [27] + Scaffold-GS [36]	28.67	0.79	0.21	0.166	0.168	0.006
MASt3R [27] + Scaffold-GS [36]*	30.92	0.90	0.13	0.047	0.103	0.005
NoPe-NeRF [5]	26.34	0.74	0.39	0.080	**0.038**	0.006
CF-3DGS [14]	31.28	0.93	0.09	0.041	0.069	0.004
Ours	**32.83**	**0.94**	**0.08**	**0.032**	0.068	**0.003**

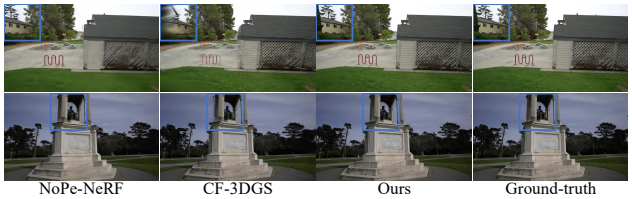


Figure 9. **Qualitative comparison on the Tanks and Temples dataset [25].** NoPe-NeRF [5] produces visibly blurred results with inaccurate geometries, while CF-3DGS [14], despite better sharpness, fails to reconstruct fine details accurately. In contrast, our LongSplat method achieves superior rendering quality, closely matching the ground truth with sharper textures, more accurate geometry, and consistent lighting.

grid size is 0.1. Camera poses are optimized via a differentiable CUDA-accelerated rasterizer, parameterized with quaternions and translation vectors. We use 400 local, 900 global, and 10,000 refinement iterations, starting with three initial frames. The octree density thresholds for splitting and removal start at 10 and 5, progressively increasing with depth. Visibility IoU threshold is set to 0.2. All experiments are conducted on a single NVIDIA RTX 4090.

4.2. Comparisons

Tanks and Temples. We evaluate LongSplat on the Tanks and Temples dataset [25], a standard benchmark for novel view synthesis. As shown in Tab. 3, LongSplat achieves state-of-the-art rendering quality (avg. PSNR: 32.83 dB) and superior camera pose estimation accuracy (lowest ATE and RPE). Qualitative results in Fig. 9 confirm sharper textures, accurate geometry, and better visual consistency compared to baselines. Please refer to the supplementary material for the full quantitative evaluation table for each scene.

Table 4. **Quantitative evaluation on the Hike dataset [39].** Our method consistently outperforms baselines across diverse scenes with complex trajectories and extended sequences, highlighting LongSplat's robustness and superior scene representation capability. CF-3DGS [14] encounters OOM in all scenes and is thus omitted.

Method	PSNR↑	SSIM↑	LPIPS↓
MASt3R [27] + Scaffold-GS [36]	17.30	0.42	0.52
MASt3R [27] + Scaffold-GS [36]*	17.90	0.44	0.50
LocalRF [39]	23.56	0.68	0.29
Ours	**25.39**	**0.81**	**0.19**

Table 5. **Ablation on training components.** Removing pose estimation, global optimization, or local optimization significantly degrades performance, highlighting each module's importance. Our full method achieves the best rendering quality and pose accuracy.

Method	PSNR↑	SSIM↑	LPIPS↓	RPE_t↓	RPE_r↓	ATE↓
w/o Pose estimation	20.19	0.56	0.51	0.42	2.71	0.71
w/o Global optimization	20.50	0.58	0.41	0.12	0.50	0.01
w/o Local optimization	25.94	0.77	0.28	0.06	0.31	0.01
w/o Refinement	26.08	0.80	0.25	0.04	0.22	0.01
Ours	**27.88**	**0.85**	**0.17**	**0.03**	**0.11**	**0.00**

Free Dataset. We evaluate LongSplat on the challenging Free dataset, achieving superior reconstruction quality as shown in Tab. 1 and Fig. 7. Competing methods like CF-3DGS often face OOM issues, while LocalRF produces fragmented geometry and pose drift. Although MASt3R + Scaffold-GS avoids OOM errors, its inaccurate global pose estimates from MASt3R result in blurred renderings and structural distortions. Our method also achieves consistently lower pose errors than baselines, as shown quantitatively in Tab. 2 and visually in Fig. 8.

Hike Dataset. We evaluate LongSplat on the challenging Hike dataset, achieving state-of-the-art reconstruction quality as shown in Tab. 4 and Fig. 10. Competing methods like CF-3DGS often fail due to OOM issue, LocalRF produces lower-quality reconstructions, and MASt3R struggles with long outdoor trajectories, resulting in poor reconstruction quality.

4.3. Ablation Studies

Training Components. To analyze the contribution of each training component, we individually disable them and evaluate performance. As shown in Tab. 5, removing pose estimation severely harms reconstruction quality and increases pose errors (ATE: 0.71). Omitting global or local optimization also reduces performance. Our full method achieves the highest quality and minimal pose errors.

Local Window Sizes. We analyze the effect of local window size on reconstruction and pose accuracy in Tab. 6. Small fixed-size windows (e.g., 1 frame) lack sufficient constraints, causing fragmentation and higher errors. Our visibility-adapted window achieves the best balance, yielding the highest reconstruction quality and lowest pose drift.

Anchor Unprojection Strategies. We compare our adaptive octree anchor formation to (1) per-pixel initialization, (2)

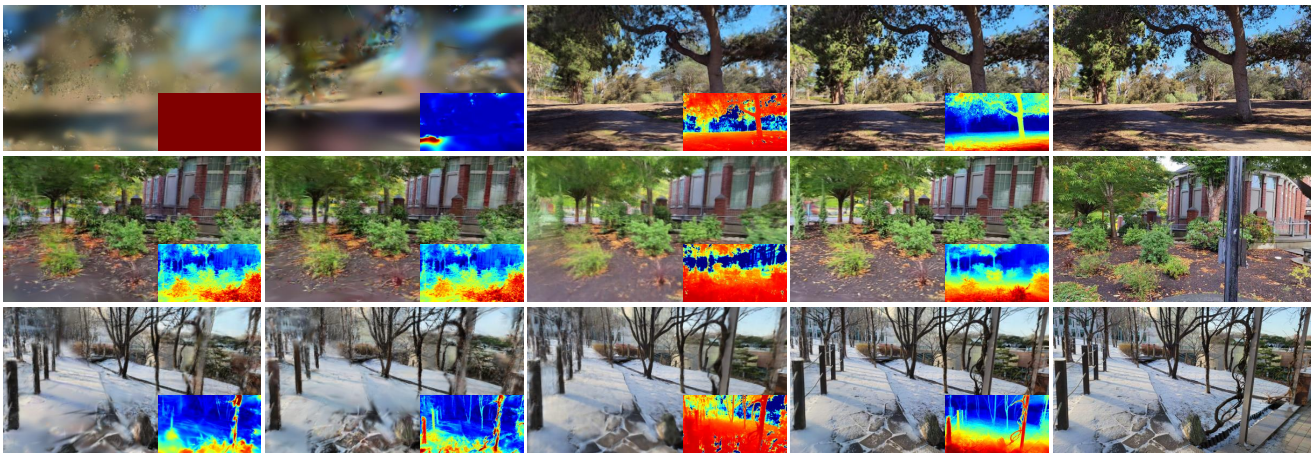


Figure 10. **Qualitative results on the Hike dataset [39].** Compared to existing methods such as LocalRF [39] and MASt3R [27]+Scaffold-GS [36], our approach significantly improves visual clarity and reconstruction fidelity, accurately capturing complex details and textures in challenging scenes captured during long, casual outdoor trajectories. Notably, our method better preserves structural details and reduces artifacts, demonstrating enhanced robustness and visual quality. "*": Initialized with MASt3R poses, then jointly optimized.

Table 6. **Ablation on local window sizes.** Fixed small windows (e.g., 1-frame or 5-frame) or global optimization degrades reconstruction quality and pose accuracy. Our visibility-adaptive window dynamically selects optimal context, achieving the best balance of local detail and global consistency.

Window size	PSNR ↑	SSIM ↑	LPIPS ↓	RPE_t↓	RPE_r↓	ATE↓
1-frame (Minimum Window)	26.58	0.80	0.23	0.05	0.21	0.01
5-frame (Fixed Window)	26.90	0.82	0.22	0.04	0.18	0.01
All Frames (Global Optimize)	26.15	0.78	0.26	0.06	0.28	0.08
Ours (Visibility-Adaptive)	**27.88**	**0.85**	**0.17**	**0.03**	**0.11**	**0.00**

Table 7. **Ablation on anchor unprojection strategies.** Our Adaptive Octree method achieves the best rendering quality and lowest perceptual errors, significantly reducing memory usage (7.92× compression) compared to baselines.

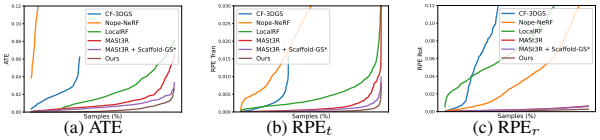
Method	PSNR↑	SSIM↑	LPIPS↓	Size (MB)↓	Compress↑
Per-pixel Unprojection (Dense)	22.47	0.69	0.35	799	1.00x
Fixed-size Voxel Unprojection	26.99	0.81	0.18	591	1.35x
Naive Densification	25.73	0.75	0.31	63	12.66x
Ours (Adaptive Octree)	27.88	0.85	0.17	101	7.92x

Table 8. **Comparison of training efficiency on the Free dataset.** Our method significantly reduces training time and achieves dramatically higher throughput (FPS) while simultaneously maintaining a compact model size compared to state-of-the-art approaches.

Method	FPS ↑	Training time ↓	Size (MB) ↓
NoPe-NeRF [5]	0.29	36 hr	7
LocalRF [39]	1.17	14 hr	1080
CF-3DGS [14]	9.81	2 hr	1966
Ours	281.71	1 hr	101

fixed-resolution voxels, and (3) naïve densification in Tab. 7. Our method achieves superior reconstruction quality with significantly reduced memory usage (7.92× compression).

Training Efficiency. We evaluate the computational efficiency of LongSplat (Tab. 8), which achieves 281.71 FPS and trains in just 1 hour on an NVIDIA RTX 4090, nearly 30× faster than LocalRF. Our method also significantly reduces the model size to approximately 101 MB.



(a) ATE (b) RPE_t (c) RPE_r

Figure 11. **Robustness analysis on camera pose estimation (Free dataset [60]).** We plot cumulative error distributions for ATE, RPE translation, and rotation. Our method consistently achieves lower errors compared to existing methods, demonstrating superior robustness and reduced pose drift.

Robustness Analysis of Camera Pose Estimation. We further analyze robustness by comparing cumulative error distributions for ATE and RPE (translation and rotation) in Fig. 11. LongSplat achieves consistently lower errors than baselines, effectively minimizing drift and maintaining stable trajectories, highlighting the advantage of our incremental optimization and robust tracking.

5. Conclusion

We present LongSplat, a robust unposed 3D Gaussian Splatting framework for casual long videos. It integrates incremental joint optimization, a robust tracking module, and adaptive octree anchors, significantly improving pose accuracy, reconstruction quality, and memory efficiency. Extensive experiments confirm that LongSplat consistently outperforms state-of-the-art approaches. Future work includes handling dynamic scenes and enhancing pose estimation robustness.

Limitations. LongSplat shares common limitations of unposed reconstruction methods, assuming static scenes and fixed intrinsics, making it unsuitable for dynamic objects or varying focal lengths.

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